

although again in routine use, present a decidedly nonphysiologic challenge. Stroke can occur through multiple mechanisms, with varying degrees of dependence on coagulation factors, platelets, and platelet-derived VWF. The experiments described by Verhenne et al, however, do document a clear role for platelet-derived VWF that is specific to ischemic stroke injury.

The insights provided by Verhenne et al raise more questions than they answer. Mice are certainly not humans, and therefore their model requires further study to understand whether or not this pathophysiology is replicated beyond the murine setting. Human and murine VWFs are also not identical. All strokes are not created equal, and platelet-derived VWF may not be relevant in all settings. However, these data do provide an intriguing base for further study of platelet-derived VWF and lend credence to its importance in particular aspects of hemostasis. Plasma VWF has received much attention, in part due to the fact that it is relatively easy to measure. It is only fair that platelet-derived VWF enjoy its role in the spotlight, and perhaps this will yield improved understanding of hemostasis and improved therapies for disease in which platelet-derived VWF plays a key role.

Conflict-of-interest disclosure: V.H.F. has served as a consultant for Baxter. ■

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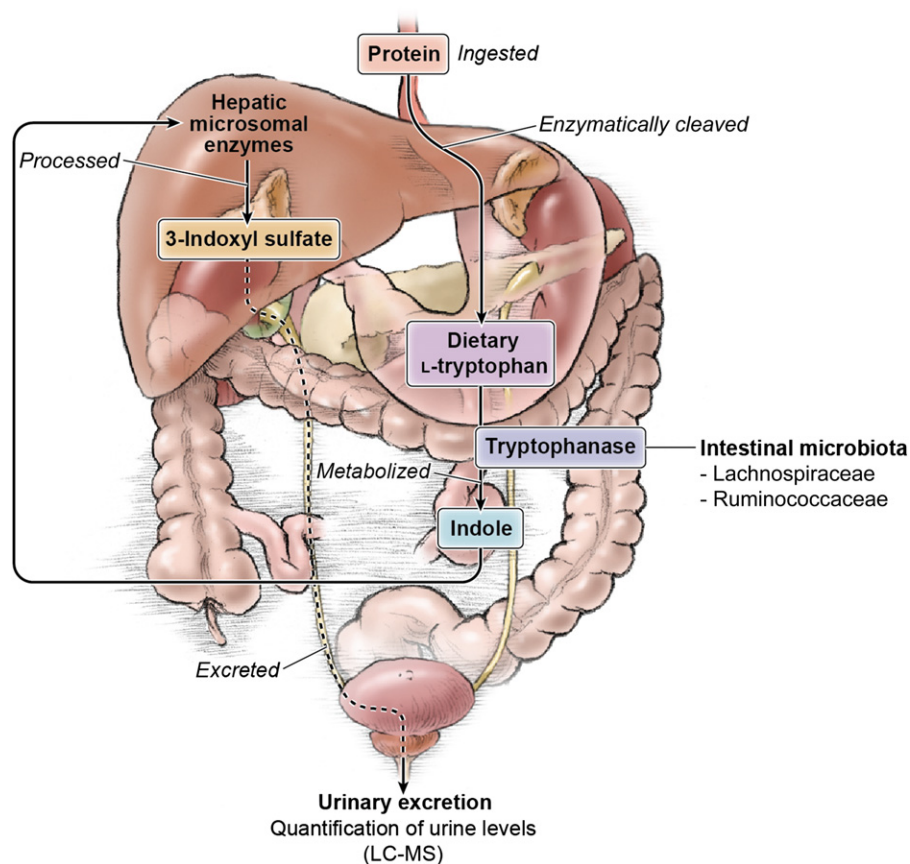
● ● ● TRANSPLANTATION

Comment on Weber et al, page 1723

How's your microbiota? Let's check your urine.

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In this issue of *Blood*, Weber and colleagues demonstrate that in the first 10 days following allogeneic hematopoietic transplantation, urinary 3-indoxyl sulfate is a biomarker of intestinal microbiota health and predicts reduced intestinal graft-versus-host disease (GVHD) and treatment-related mortality, as well as improved overall survival.¹



Urinary 3-indoxyl sulfate generation. Orally ingested proteins are enzymatically cleaved to produce tryptophan, which is then metabolized by tryptophanase-expressing intestinal bacteria into indole and absorbed by the intestinal tract. Microsomal enzymes in the liver then process indole into 3-indoxyl sulfate, which is excreted into the urine and can be quantified by liquid chromatography-mass spectrometry (LC-MS). Professional illustration by Ken Probst, Xavier Studio.

With the increased availability and reduced costs of next-generation deep sequencing modalities, nucleotide-based analysis of complex populations, including intestinal bacteria, has become an increasingly commonly used assay. Studies have uncovered surprising relationships between intestinal commensal bacterial communities and various aspects of human health and disease. GVHD, a common and often severe inflammatory syndrome that occurs in patients undergoing allogeneic hematopoietic transplant, has recently been found by several groups to be closely associated with disruptions in the normal community structure of intestinal commensal bacteria.²⁻⁵ More recently, studies have found that microbiota injury frequently precedes severe GVHD, raising the possibility that loss of protective commensal bacteria during the course of the transplant may contribute to the development of GVHD pathophysiology.⁵⁻⁷

Clinical strategies to intervene when microbiota injury occurs are being developed. To identify high-risk patients who could benefit most, a quick method for determining the “health” of an individual’s microbiota would be an invaluable tool. Unfortunately, routine rapid turnaround of deep-sequencing results is currently not feasible for clinical use; for deep sequencing to be affordable, samples need to be batched, often hundreds of samples at a time.

Quantitative polymerase chain reaction using primers specific for bacterial subsets is one approach to quickly assay the intestinal microbiota. The gene most commonly targeted encodes for 16S ribosomal RNA, where regions are conserved between different species of bacteria, allowing for phylogenetic classification. Primers have been designed

to quantify the 16S ribosomal RNA genes of specific beneficial bacterial subsets.⁸ Alternatively, primers can be designed to quantify other genes expressed by beneficial bacterial subsets; for example, butyryl-coenzyme A transferase is expressed by many fermentative bacteria capable of producing butyrate, which mediates anti-inflammatory effects in the colon.⁹

Weber and colleagues have now identified a novel alternative strategy of quickly assessing microbiota health, via quantifying urinary 3-indoxyl sulfate. This assay capitalizes on the expression of tryptophanase by subsets of the intestinal microbiota. Dietary L-tryptophan is converted by tryptophanase to indole, which is absorbed and metabolized by the liver into 3-indoxyl sulfate and then excreted in the urine (see figure). Weber and colleagues have developed a liquid chromatography–mass spectrometry assay to quantify urinary 3-indoxyl sulfate levels. They previously reported that 3-indoxyl sulfate levels are decreased in the setting of GVHD-associated microbiota injury.⁵ Now they have advanced their previous findings by demonstrating that 3-indoxyl sulfate levels are associated with specific subsets of the intestinal microbiota (Lachnospiraceae and Ruminococcaceae), and not with others (Bacilli). Notably, bacterial subsets associated with high 3-indoxyl sulfate levels have been linked to reduced intestinal inflammation. Thus, the authors asked whether quantifying levels of this urinary biomarker early after transplant (within 10 days) can predict later outcomes. Indeed, patients with low 3-indoxyl sulfate levels, indicating a damaged intestinal microbiota, developed more transplant-related mortality and had reduced overall survival.

Whether indole and its metabolites may be biologically active and mediate anti-inflammatory effects in the setting of GVHD is an important point raised by this study. Further study, likely in animal models, could shed light on this question.

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